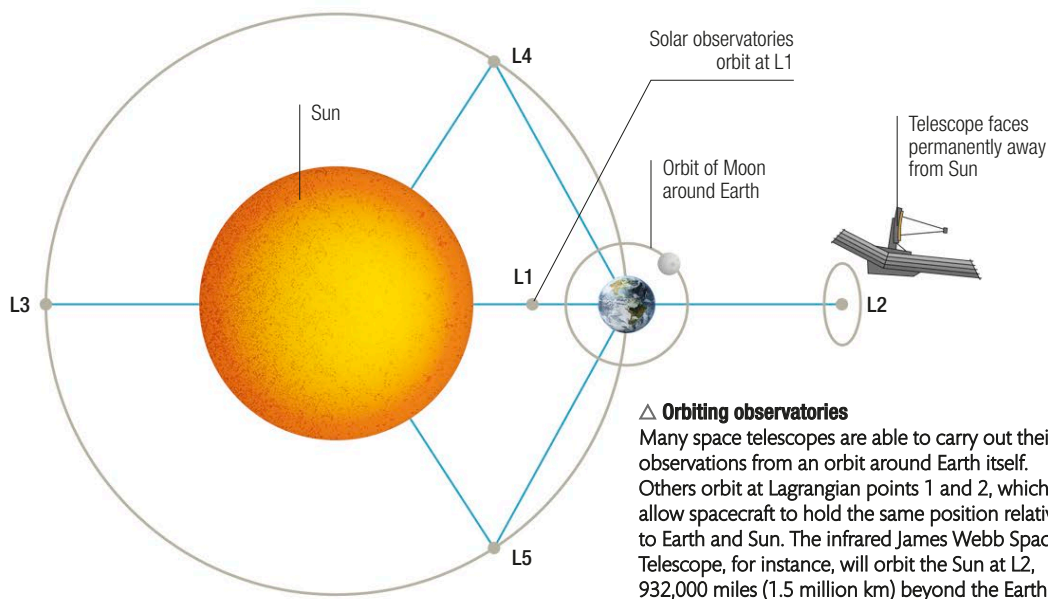
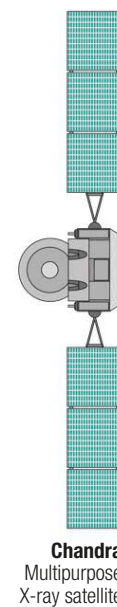
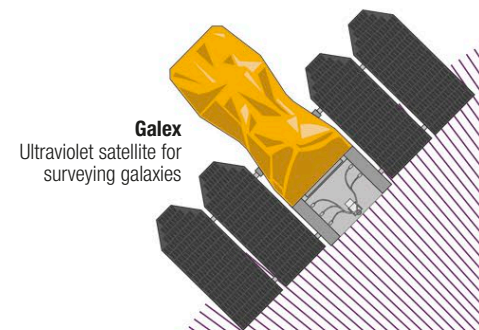
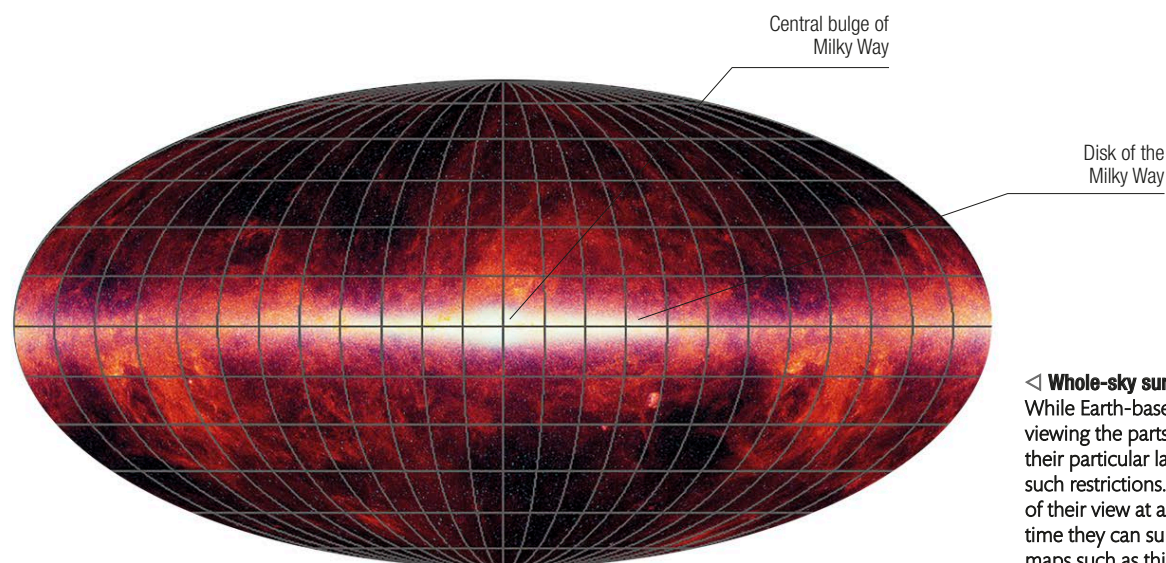




SPACE TELESCOPES

TELESCOPES ORBITING ABOVE EARTH OPEN UP NEW VISTAS ON THE UNIVERSE. THEY NOT ONLY REVEAL OBJECTS WHOSE INVISIBLE RADIATIONS ARE BLOCKED BY OUR PLANET'S ATMOSPHERE BUT ALSO ALLOW ASTRONOMERS TO CARRY OUT MORE PRECISE STUDIES IN VISIBLE LIGHT.

The visible light with which we view the Universe is simply a type of electromagnetic wave. These are packets of energy that move through space in the form of self-reinforcing electric and magnetic fields. The properties of light we perceive as colors depend on the wavelength of these waves, but the broader electromagnetic spectrum goes far beyond them to encompass much longer and shorter waves. Robotic space telescopes allow astronomers to study these elusive wavelengths, but they are often very different from Earth-bound instruments. For example, infrared telescopes need extreme cooling so that their own heat does not swamp the weak signals, while X-rays and gamma rays will pass straight through most traditional mirrored telescope designs.



X-rays have wavelengths from 0.01 to 10 nanometers (nm). They are emitted from superhot objects such as disks around black holes and gas in galaxy clusters.

X-RAYS

Fermi
Telescope to study mysterious gamma-ray bursts

◀ Whole-sky surveys

While Earth-based telescopes are limited to viewing the parts of the sky that are visible from their particular latitude, space telescopes have no such restrictions. The Earth may block a large part of their view at any moment, but with enough time they can survey the entire sky and build up maps such as this one at infrared wavelengths